Checkout Incentives & Customer Lifetime Value

An end-to-end data analysis & machine learning project on the Brazilian Olist E-Commerce Dataset

Data Audit | LTV Modeling | A/B Testing Framework | Incentive Economics | Category Analysis | Predictive Modeling

This project analyzes e-commerce customer and order data to answer both technical data-analysis questions and business questions around customers, orders, revenue, retention, checkout incentives, and purchasing behavior. It progresses from a structured data audit through customer-value modeling, incentive economics, category-level retention analysis, and a machine-learning approach to repeat-purchase prediction.

96,096	$16.01M	3.12%	45,432	75.18%
Unique Customers	Total Historical Revenue	Repeat-Customer Rate	High-Value One-Time Targets	Revenue from Target Segment

01 · Data Audit

Nine interrelated tables (Customers, Geolocation, Orders, Order Items, Payments, Reviews, Products, Sellers, Category Translation) covering 99,441 order-linked customer records and 96,478 delivered orders were audited for structure, missingness, duplication, chronology, and financial consistency.

- Customer→order relationships: **complete** (0 orphan records).
- Payment→order linkage: complete except **1 order** (30710).
- 98,284 of 98,665 orders (99.61%) show exact item/payment financial matches; 381 discrepancies flagged (mean $8.58, max $182.81).
- 166 shipment-before-purchase and 23 delivery-before-shipment records flagged as chronological anomalies.
- Geolocation contains 261,831 exact duplicates of 1,000,163 rows; 17,781 ZIP prefixes show unstable coordinates.
- `customer_id` (99,441) ≠ `customer_unique_id` (96,096) — the latter is required for correct LTV computation.
- SP, RJ and MG dominate customer/order activity (SP: 40,302 customers, $5.20M sales).
- No records were deleted; anomalies were flagged, not removed, to preserve data fidelity.

02 · Customer Lifetime Value (LTV) Modeling

Historical LTV (total payment value per `customer_unique_id`) was computed for 96,096 customers, totaling **$16,008,872.12**. Mean LTV ($166.59) exceeds median LTV ($108.00), confirming a right-skewed, Pareto-style value distribution.

Segment	Customers	Share	Avg LTV	Median LTV	Revenue Share
One-time buyers	93,099	96.88%	$161.82	$105.70	94.10%
Repeat buyers	2,997	3.12%	$314.99	$225.84	5.90%
High-value one-time (target)	**45,432**	**47.28%**	**$264.91**	**$180.21**	**75.18%**

Revenue concentration: Top 1% of customers → 10.48% of revenue · Top 10% → 38.51% · Top 20% → 53.77%. Repeat customers show ~1.95× higher mean LTV and ~2.14× higher median LTV than one-time buyers. 36.60% of repeat customers return within 7 days; 51.1% within 30 days.

03–04 · Checkout Incentive & ROI Analysis

The **45,432 high-value one-time customers** are the primary incentive target — 47.28% of customers generating 75.18% of revenue, average LTV $264.91. Notably, repeat rate *declines* with first-order value (3.41% for orders <$50 vs 2.03% for orders ≥$1,000), so high-value customers are targeted for economic importance, not higher repeat propensity. Delivery time (12.57 vs 12.39 days) and review score (4.15 vs 4.16) do not meaningfully differentiate repeat from one-time customers.

Incentive / Conversion	Avg 2nd-Order Value	Est. Revenue (5%)	Incentive Cost	Net Revenue	ROI
$25 incentive, 5% conversion	$148.51	$337,423.52	$56,800	$280,623.52	494.06%
$25 incentive, 1% conversion	$148.51	$67,425.30	$11,350	$56,075.30	494.06%
$50 incentive (any rate)	$148.51	—	—	—	197.03%
$100 incentive (any rate)	$148.51	—	—	—	48.51%

ROI is constant across conversion rates for a fixed incentive (revenue & cost scale linearly); conversion rate sets the *scale* of opportunity, incentive size sets the *economics per conversion*. Recommendation: run a controlled A/B test (control vs. $25 treatment) and measure incremental conversion, not raw conversion.

05 · Product Category Retention Analysis

112,650 order-items across 72 categories were analyzed against an overall repeat-item baseline of **6.72%**. Using repeat-share lift, statistical significance (Mann-Whitney U + Benjamini-Hochberg FDR correction), and a $50,000 revenue floor, five categories were classified **High Priority**:

Category	Repeat Share	Lift vs Baseline	Revenue	Revenue Share
Bed, Bath & Table	10.02%	+3.30 pp	$1,241,681.72	7.84%
Computers & Accessories	7.05%	+0.33 pp	$1,059,272.40	6.69%
Furniture Decor	9.85%	+3.13 pp	$902,511.79	5.70%
Fashion Bags & Accessories	12.95%	+6.23 pp	$184,273.54	1.16%
Home Appliances	17.90%	+11.18 pp	$94,990.43	0.60%

Together these five categories account for 30,078 items, 2,888 repeat items (9.60% repeat share), and **21.98% of total category revenue ($3.48M)**. Key nuance: the category with the highest repeat propensity is not the highest-value one — Home Appliances has the strongest lift (+11.18 pp) but repeat-customer item value is 37.91% lower than one-time; Bed, Bath & Table combines strong repeat behavior with the largest revenue base and only a 4.89% value gap.

06 · Repeat-Purchase Prediction (Machine Learning)

A supervised pipeline (Logistic Regression, Random Forest, XGBoost, LightGBM, CatBoost, AdaBoost) was trained on 93,253 customers using first-order features (items, product count, merchandise/freight/total value, purchase month/day/hour, state) to rank customers by repeat-purchase likelihood — a severely imbalanced problem (repeat rate 3.08%).

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
CatBoost (primary)	**61.21%**	**3.84%**	**48.17%**	**0.0711**	**0.5770**	**0.0432**
XGBoost (alt.)	**62.48%**	**3.97%**	**48.17%**	**0.0734**	**0.5643**	**0.0424**
Random Forest	93.21%	4.36%	5.74%	0.0495	0.5631	0.0397
AdaBoost (rejected)	96.92%	0.00%	0.00%	0.0000	0.5282	0.0332

High accuracy is misleading under class imbalance: AdaBoost hits 96.92% accuracy while detecting *zero* repeat customers. At Top-10% targeting, CatBoost captures 15.48% of repeat customers (1.548× lift over random); the simple "first-order item count" baseline alone already achieves 1.461× lift, showing purchase quantity is a strong, explainable early signal. Six features remained significant after FDR correction: first-order items, product count, merchandise value, total value, purchase month, and purchase hour.

Business Recommendation & Key Takeaways

- **Target, don't blanket-discount:** prioritize the 45,432 high-value one-time customers (75.18% of revenue).
- **Keep incentives well below $148.51** (observed avg. second-order value) — a $25 reference incentive yields ~494% modeled ROI.
- **Multi-window engagement:** repeat purchases occur from same-day to 180+ days out — no single timing window captures the full opportunity.
- **Layer in category targeting:** Bed/Bath/Table, Computers, Furniture Decor, Fashion Bags, and Home Appliances show the strongest repeat signal + revenue combination.
- **Use ML for ranking, not classification:** CatBoost/XGBoost meaningfully outperform random targeting but should prioritize (Top-K), not gate, campaign eligibility.
- **Validate causally:** all findings are observational — a controlled A/B test (control vs. treatment) is required before scaling any incentive program.